

Modeling and Predicting Behaviors of Inductive Traffic Loops

Cooper Amador-Risch, **Tushar Mehta**, Swetha Menon, Project TA Jack Barsotti, Project Advisor Dr. Chayanika Roy Chaudhuri

Department of Physics, Mellon College of Science.

Introduction

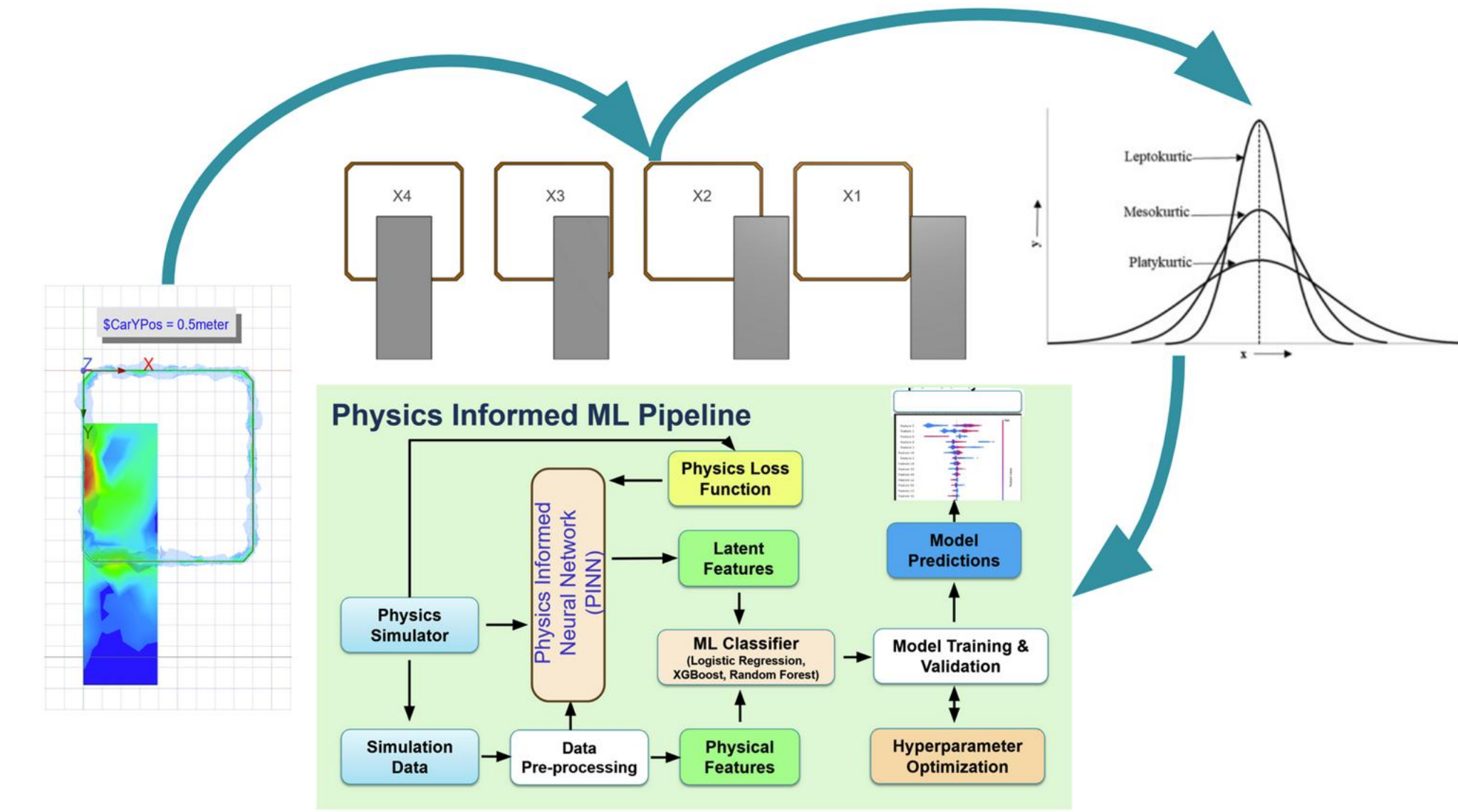
Modern transportation systems increasingly rely on real-time data to manage traffic flow, ensure safety, and infrastructure planning. Among the most widely adopted tools in this domain are inductive traffic loop detectors; electromagnetic sensors embedded in roadways that detect the presence of vehicles. The detectors are foundational data for intelligent transport systems, enabling applications such as dynamic signal control, congestion monitoring, and traffic management.

Inductive traffic looping refers to the strategic placement and use of inductive loops to monitor traffic at critical junctions and road segments. When a vehicle passes above the loop, it induces a change in inductance. This induced change is detected by roadside electronics and translated into actionable data. Inductive loops are used widely, but are not utilized to gain the most amount of traffic data like vehicle positioning and type. Our goals were to:

- Accurately simulate a vehicle passing over an inductive traffic loop in Ansys HFSS
- Use simulation and statistical techniques to get data of a motorcycle passing over different positions of the loop.
- Utilize Trained Models to detect the position of a motorcycle in one of four inter-lane positions

Methods

1. Model the motorcycle in HFSS using realistic dimensions. Calculate the inductance value for 6 time steps
2. Run simulations for the motorcycle at each of the 4 positions (x1, x2, x3, x4)
3. Use that data to distribute it along a probability distribution function in order to have many data points of inherent randomness
4. Train a model on generated data to predict the most likely lane position of a motorcycle at a traffic stop.



Data & Results

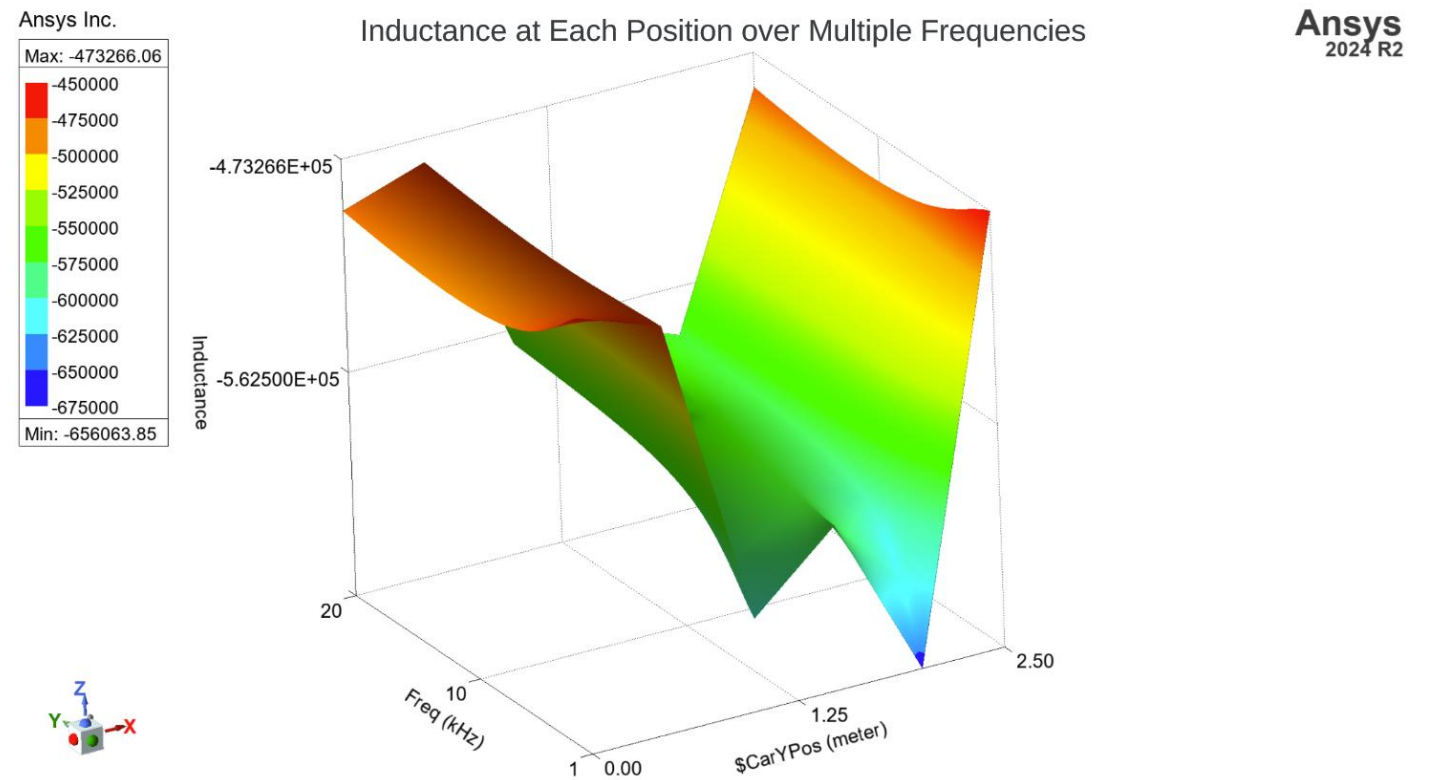


Fig 1. Inductance values as the vehicle passes over the traffic loop when the loop operates at different frequencies. Shows that loop behaves similarly at multiple operating frequencies.



Fig 2. A) Data generation with Gaussian Noise and Scale Factor, very dispersed and overlapping. B) Data generation with only Gaussian Noise, less dispersed but still overlapping. C) Data generation with only Hyperbolic Secant Noise, less dispersed and less overlapping.

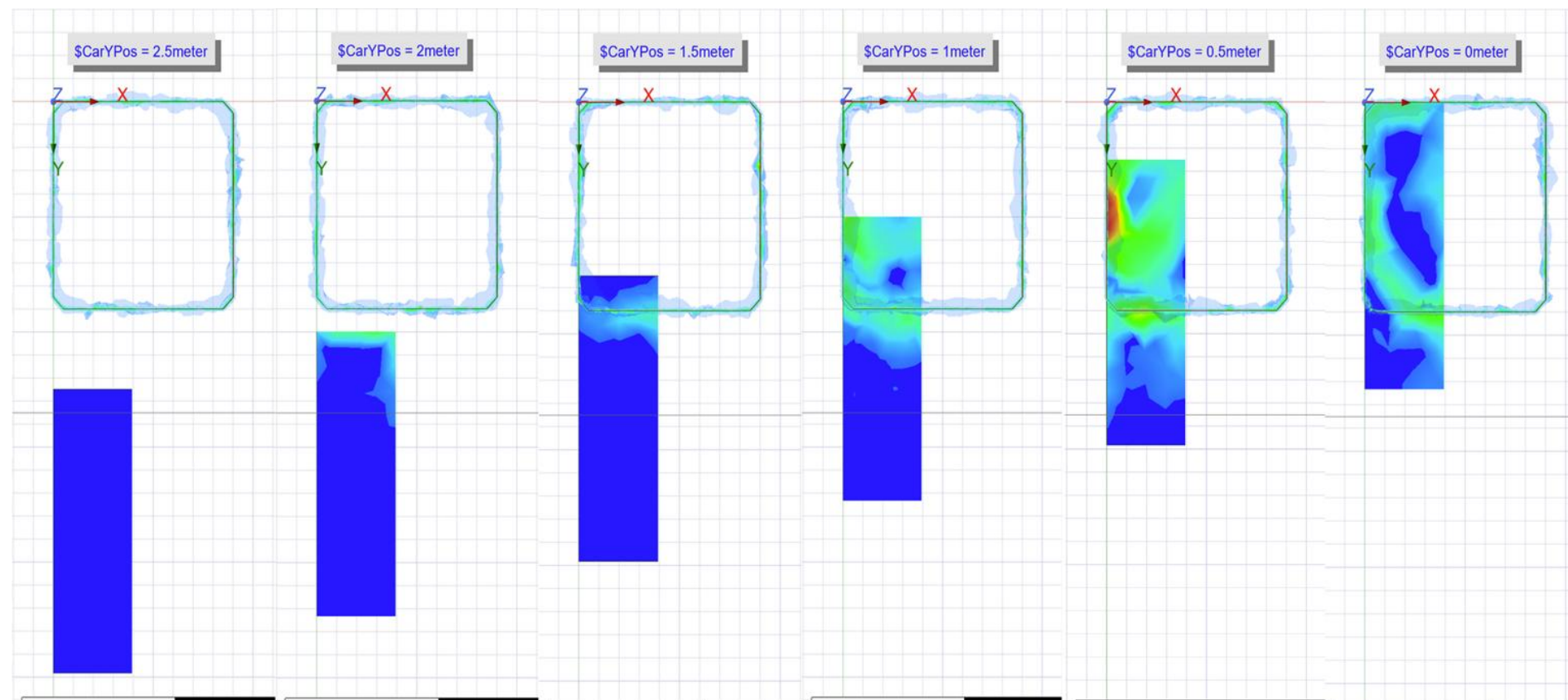


Fig 3. Example of Simulations as the motorcycle moves over the inductive loop. This displays the electric field of the loop and the magnetic field of the motorcycle.

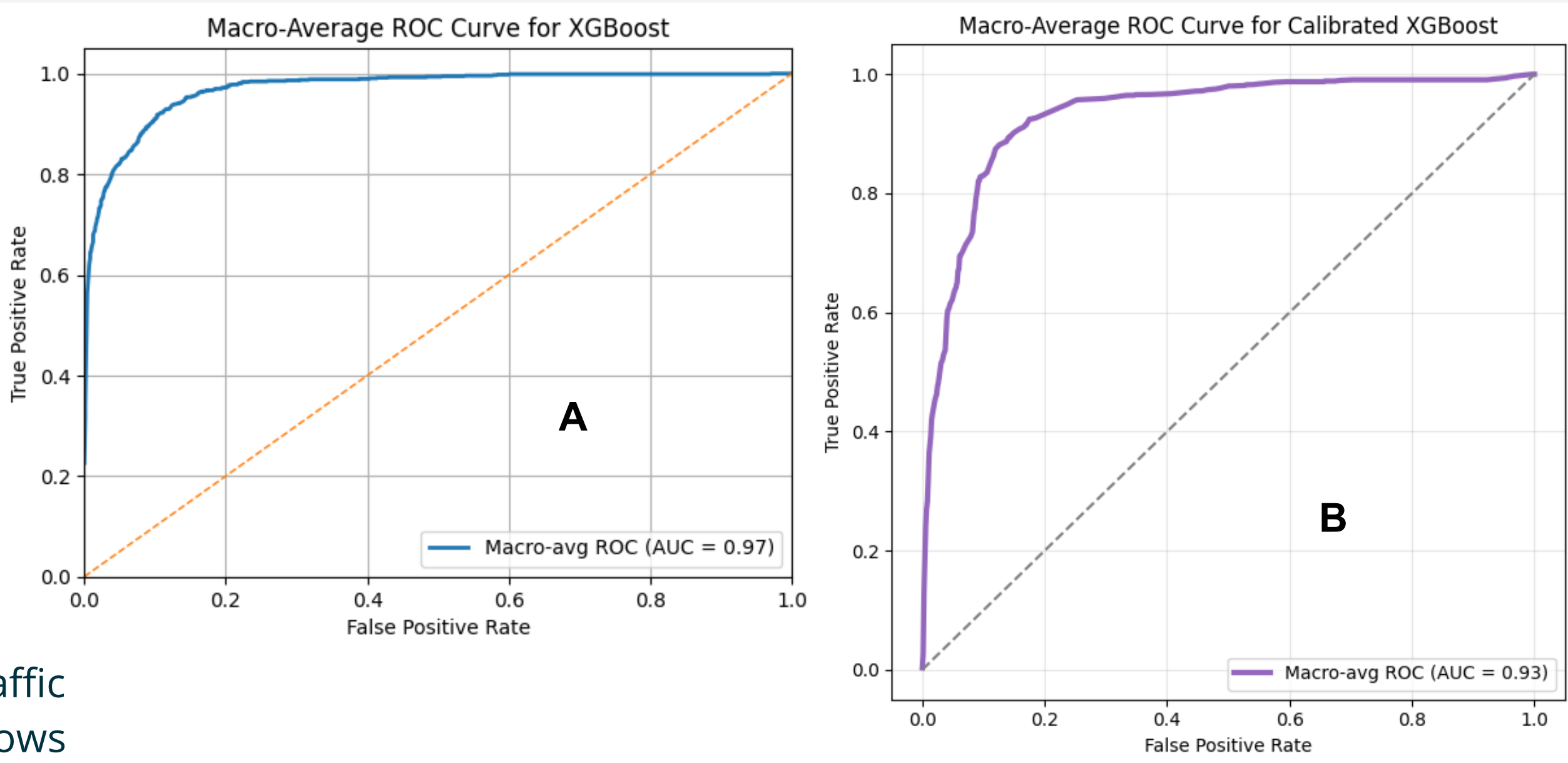


Fig 4. A) ROC Curve for XGBoost classification with All Features of Physics Informed Neural Network (PINN). B) ROC Curve for XGBoost classification with only Inductance for Neural Net (NN). Classification with All Features for PINN is more accurate than only inductance as ROC Curve for A has more area under the curve than B.

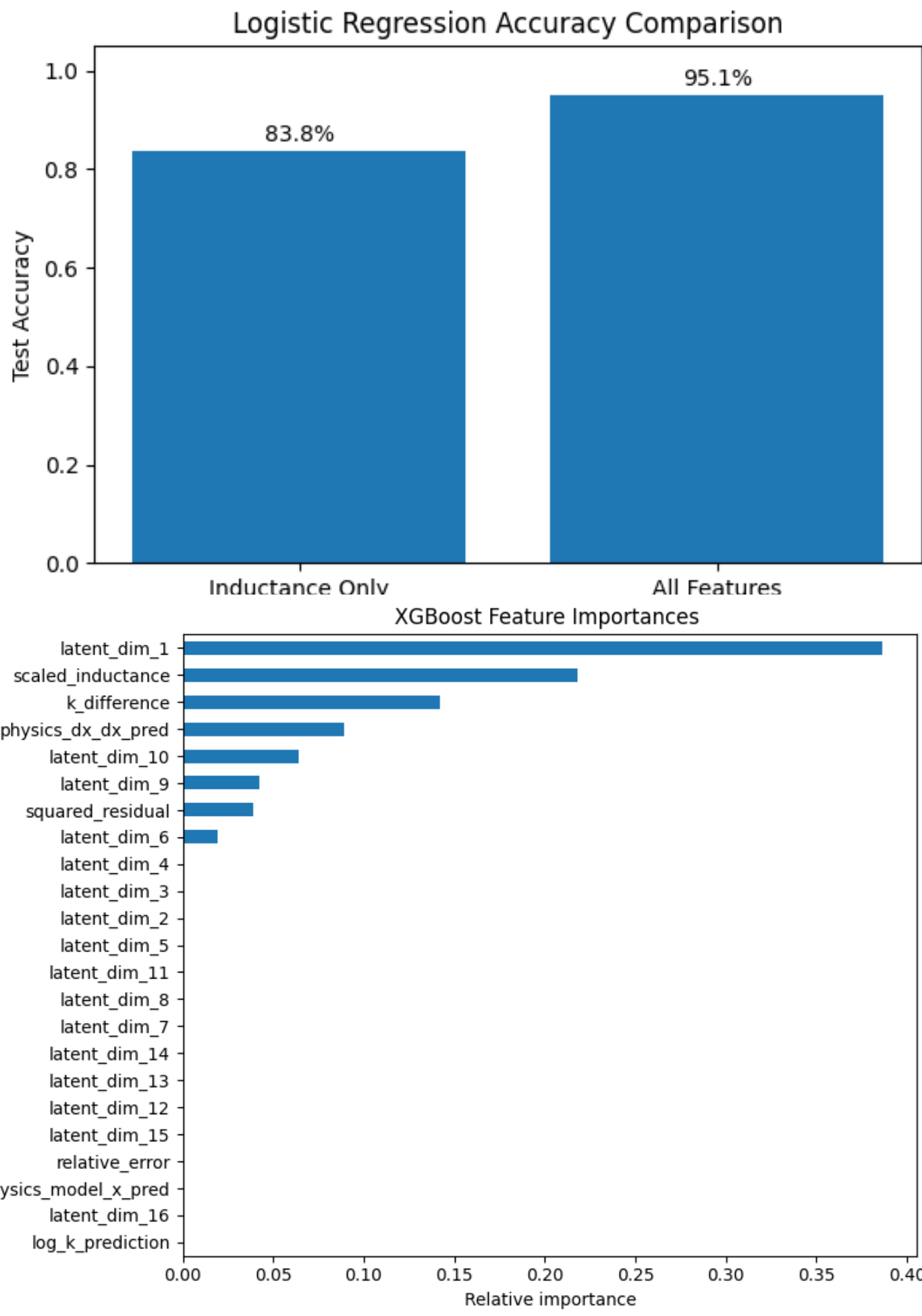


Fig 5. Accuracy Comparison for Logistic Regression classification between All Feature PINN and Inductance only NN. All Feature PINN has 95.1% accuracy and Inductance only NN has 83.8% accuracy, making All Feature PINN more accurate.

Fig 6. Feature Importance for XGBoost classification. Shows how important each feature is to the All Feature PINN

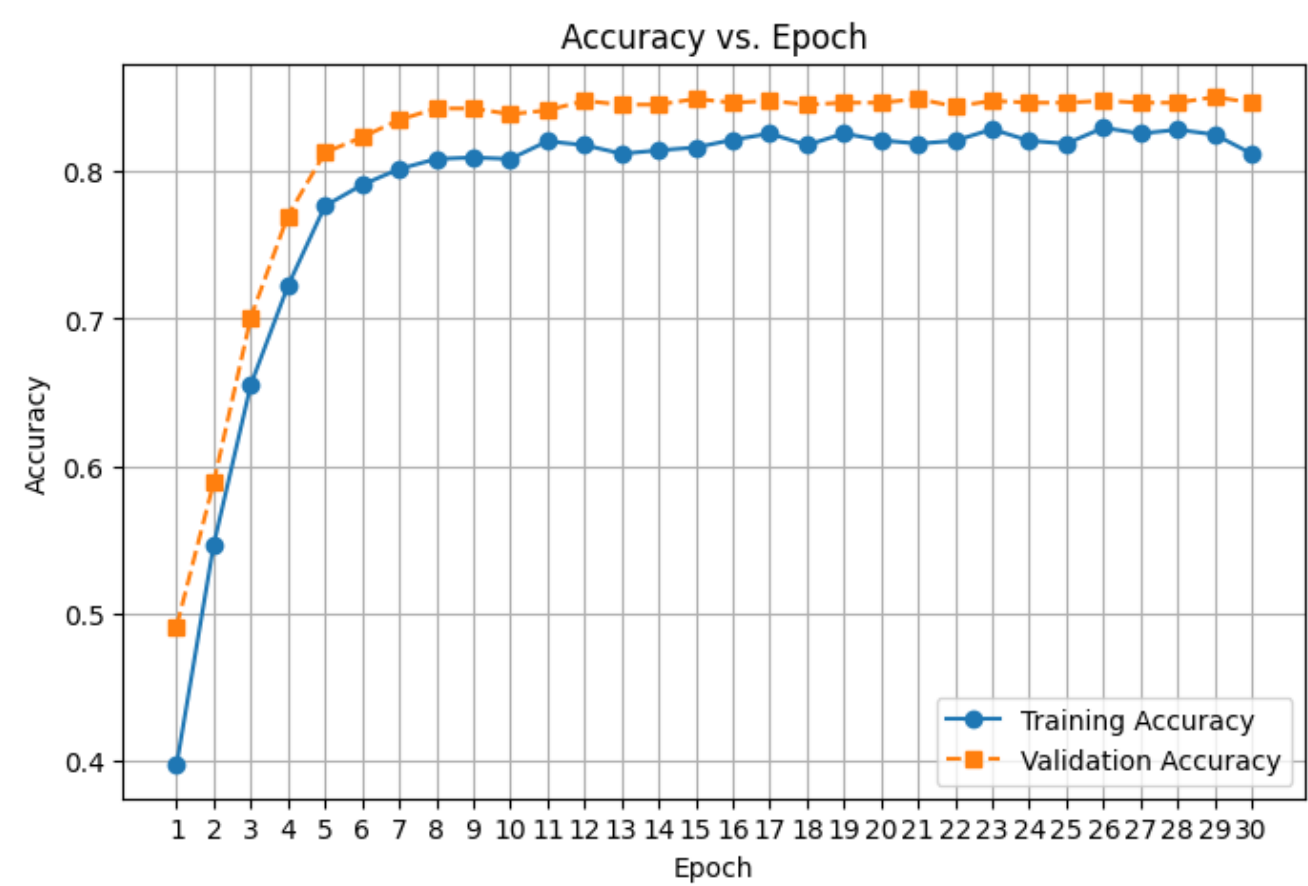


Fig 7. Training of Early Neural Net Model. Only reached mid 80% accuracy before using PINN to get above 95% accuracy.

Conclusion

- Hyperbolic Secant Noise was better at modeling data for the system than Gaussian Noise due to its tighter distribution
- The goal of accurately detecting the position of the motorcycle was achieved
- The model with the addition features(Physical Features + PINN features) outperformed the traditional linear regression model.
- The first feature from the PINN was more important to the classification than the Induction was

Based on the outcomes of this testing, it can be concluded that a real world model of an inductive traffic loop can accurately detect the position of a motorcycle within a lane by using a Physics Informed Neural Network (PINN)

Future Directions

- Gather more simulation data
 - Is the model accurate?
- Vehicle Type Detection
- Speed and Acceleration Modeling
- Different Geometries
 - Utilizing Optimization Techniques
 - Better Detection?
- Using Machine Learning for finding both x & y position
- Using multiple traffic loops chained in series
 - Better detection capabilities?

Acknowledgements

We would like to acknowledge our team project advisors Dr. Chaudhuri and Jack Barsotti for their support throughout this project. Thanks to John Urbanic from PSC for exposing us to PINNs. We would also like to thank Lane Olson for her support.

We would like to thank the **PGSS Campaign, Inc** for their generous efforts to secure funding for the PGSS Program. We also thank the **PA Department of Education** for their continued support of PGSS.

